Lewis Thompson

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Sunshine Coast, QLD — Brisbane-commutable

Product engineer. Shipped a native iOS health app solo and runs its production infrastructure; builds document-grounded LLM systems. Everything built with supervised AI coding agents.

EXPERIENCE

Founder & Sole Engineer — Pum (pum.me) Jul 2025–present

- Designed, built and shipped an iOS endometriosis tracker (App Store, Jul 2026), now #4 in US App Store search for "endometriosis tracker". Its doctor's report follows NICE guideline NG73.
- Built with supervised Claude Code agents working in parallel on separate git worktrees. I define the goal and how it is verified, and any check I keep repeating becomes a test that fails the build.
- Wrote the core as a Rust engine (cycle maths, statistical insight models) bridged to SwiftUI with UniFFI; it must give Pum and its companion app identical results, and its numerics match published references.
- A new insight model (n-of-1, case-crossover) replaces the baseline only if an eval harness shows it finds more real effects in simulated users, with a false-positive rate of 5% or less.
- Analysed ~11,800 rival app reviews with my own Rust and LLM tool: 180 asked for partner sharing, and the one app with it churned least (7.5% vs Flo 15.6%). That became Circle, Pum's partner app.
- Tracks retention with private telemetry (no symptom data, IPs or device IDs): D1/D7/D30 vs a preset threshold.
- Operate the production servers: Rust (Axum) and SQLite behind nginx on DigitalOcean.

Sole Engineer — MAGI, document-intelligence platform (projectmagi.app) May 2026–present

- Document-grounded LLM pipeline (OCR, digests, retrieval) over the US National Archives UAP disclosure record: ~9,000 pages live of ~116,000. Rust (Axum) API, Postgres/pgvector, React reader, public JSON API.
- MCP server (Rust) so users can explore the archive alongside their AI assistant, which reads it through tools. Every result cites its page and span and names the pipeline run behind it.
- The model must quote the document word for word, then code finds that quote on the page. Claims that can't be found are refused and every statement links to the exact passage in the original file.
- Every model and prompt is picked by benchmark. OCR: 9 engines on 46 hard pages against a blind answer key; the winner cut character error from 32% to 15% (~$34 for all of Project Blue Book). Splitting bundled scans: vision models found 97.2% of document breaks vs 84% for text rules.
- Three repeats per setting overturned a one-run result: reasoning mode's real gain was quote accuracy (92% of quotes found on the page vs 79%). The model that grades outputs never produces them.
- Kept the UI from looking AI-generated: agents are banned from its tells (eyebrow labels, captions everywhere, needless borders, stock spacing) and design from moodboard images, guided by what I name as good.

Other work

- **Seafaring map**: did the monsoon decide where Bronze Age ports grew on the Arabian Sea? Simulated drifting and sailing boats on 28 years of wind and current data, ranked 499 coast segments by network position, and tested them against carbon-dated sites in a pre-registered permutation test.
- **Freelance Solidity smart-contract auditor** (2022–2023): protocol audits and bug bounties.
- **Noosa Cane Juice** (2022–present, part-time): market stall sales and operations.

STACK

Engineering: Swift/SwiftUI, Rust (Axum), Python, TypeScript/React, Postgres/pgvector, SQLite, Linux, nginx.

AI: Claude Code; Anthropic, OpenAI and DeepSeek APIs; embeddings and retrieval; LLM evals; MCP servers.

EDUCATION

University of Queensland — studies in mathematics and law

Full-time, available immediately · Referees on request